

COS203 MASTER REVISION BANK

DISCRETE MATHEMATICS — COMPLETE REVISION

Questions 1-300 (With Answers + Solutions)

Question 1

Let:

$$A = \{1, 2, 3, 4, 5\}$$

How many subsets does A have?

- A. 5
- B. 10
- C. 25
- D. 32

Answer: ✓ D. 32

Solution:

The number of subsets of a set with n elements is 2^n .

Here: $n = 5 \rightarrow 2^5 = 32$

So: $|P(A)| = 32$

Exam Trap:

Do not use n^2 or $2n$. The power set formula is always 2^n .

Question 2

If: $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$

Find: $A \cup B$

- A. $\{1, 2, 3\}$
- B. $\{3\}$
- C. $\{1, 2, 3, 4, 5\}$
- D. $\{4, 5\}$

Answer: ✓ C. $\{1, 2, 3, 4, 5\}$

Solution:

Union means combine all elements without repetition. $A \cup B = \{1, 2, 3, 4, 5\}$

Question 3

Using the same sets: $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

Find: $A \cap B$

- A. $\{1, 2\}$
- B. $\{3\}$
- C. $\{4, 5\}$
- D. $\emptyset$

Answer: ✓ B. $\{3\}$

Solution:

Intersection means elements common to both sets. Common element: $3 \rightarrow A \cap B = \{3\}$

Question 4

If: $|A| = 20$, $|B| = 15$, $|A \cap B| = 5$

Find: $|A \cup B|$

- A. 40
- B. 35
- C. 30
- D. 25

Answer: ✓ C. 30

Solution:

Formula: $|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 5 = 30$

Question 5

A set containing no element is called:

- A. Universal set
- B. Empty set
- C. Power set
- D. Infinite set

Answer: ✓ B. Empty set

Explanation:

The empty set is written as $\emptyset$ and contains zero elements.

Question 6

If: $A = \{a, b, c\}$

How many elements are in the power set of A?

- A. 3
- B. 6
- C. 8
- D. 9

Answer: ✓ C. 8

Solution:

Number of elements: $n = 3$. Power set: $2^3 = 8$

Question 7

Which of the following is a subset of: $A = \{1, 2, 3, 4\}$

- A. $\{1, 5\}$
- B. $\{2, 4\}$
- C. $\{0, 1\}$
- D. $\{5, 6\}$

Answer: ✓ B. $\{2, 4\}$

Explanation:

Every element of the subset must exist inside A.

Question 8

If: $A \subseteq B$ then:

- A. Every element of B is in A
- B. Every element of A is in B
- C. A and B must be equal
- D. A has more elements than B

Answer: ✓ B

Explanation:

$A \subseteq B$ means all elements of A are contained in B.

Question 9

Find: $\{1, 2, 3\} \times \{a, b\}$

- A. $\{(1, a), (2, b)\}$
- B. $\{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$
- C. $\{1, 2, 3, a, b\}$
- D. $\{(a, 1), (b, 2)\}$

Answer: ✓ B

Solution:

Cartesian product pairs every element of the first set with every element of the second.

Question 10

If: $A = \{1, 2, 3\}$

How many ordered pairs are in: $A \times A$

- A. 3
- B. 6
- C. 9
- D. 12

Answer: ✓ C. 9

Solution:

Formula: $|A \times A| = |A| \times |A| = 3 \times 3 = 9$

Batch 1 (Continued): Questions 11-25

Question 11

Given: $A = \{1, 2, 3, 4, 5\}$

Find: $A - \{2, 4\}$

- A. $\{2, 4\}$
- B. $\{1, 3, 5\}$
- C. $\{1, 2, 3, 4, 5\}$
- D. $\emptyset$

Answer: ✓ B. $\{1, 3, 5\}$

Solution:

Set difference means remove the elements of the second set from the first.

Remove 2, 4 from $\{1, 2, 3, 4, 5\}$. Remaining: $\{1, 3, 5\}$

Question 12

If: $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $A = \{2, 4, 6, 8\}$

Find A' (complement of A).

- A. $\{1, 3, 5, 7\}$
- B. $\{2, 4, 6, 8\}$
- C. $\{1, 2, 3, 4\}$
- D. $\emptyset$

Answer: ✓ A. $\{1, 3, 5, 7\}$

Solution:

Complement: $A' = U - A$. Remove $\{2, 4, 6, 8\}$ from U. Remaining: $\{1, 3, 5, 7\}$

Question 13

Which of the following represents De Morgan's law?

- A. $(A \cup B)' = A' \cup B'$
- B. $(A \cup B)' = A' \cap B'$
- C. $(A \cap B)' = A' \cap B'$
- D. $A \cup A = A'$

Answer: ✓ B

Explanation:

De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$

Question 14

If: $A \cap B = \emptyset$ then A and B are called:

- A. Equal sets
- B. Universal sets
- C. Disjoint sets
- D. Infinite sets

Answer: ✓ C. Disjoint sets

Explanation:

Disjoint sets have no common elements. Therefore $A \cap B = \emptyset$.

Question 15

In a class of 100 students:

- 60 study Java
- 50 study Python
- 20 study both

How many study at least one of the two courses?

- A. 90
- B. 110
- C. 80
- D. 70

Answer: ✓ A. 90

Solution:

Formula: $|A \cup B| = |A| + |B| - |A \cap B| = 60 + 50 - 20 = 90$

Question 16

A relation from set A to set B is a subset of:

- A. $A + B$
- B. $A \times B$
- C. $A \cap B$
- D. $A - B$

Answer: ✓ B. $A \times B$

Explanation:

A relation consists of ordered pairs. Ordered pairs come from the Cartesian product $A \times B$.

Question 17

Given: $R = \{(1,1), (2,2), (3,3)\}$ on set $A = \{1, 2, 3\}$

The relation is:

- A. Reflexive
- B. Symmetric only
- C. Transitive only
- D. None

Answer: ✓ A. Reflexive

Solution:

A relation is reflexive if every element relates to itself. Required pairs (1,1), (2,2), (3,3) are all present → Reflexive.

Question 18

A relation R is symmetric if:

- A. If aRb , then bRa
- B. If aRb , then aRa
- C. If aRb and bRc , then aRc
- D. Every element has a partner

Answer: ✓ A

Explanation:

Symmetric property: $aRb \Rightarrow bRa$

Question 19

Which property requires: aRb and $bRc \Rightarrow aRc$

- A. Reflexive
- B. Symmetric
- C. Transitive
- D. Antisymmetric

Answer: ✓ C. Transitive

Question 20

Let: $R = \{(1,2), (2,3)\}$

For R to become transitive, which pair must be added?

- A. (2, 1)
- B. (3, 2)
- C. (1, 3)
- D. (3, 1)

Answer: ✓ C. (1, 3)

Solution:

Given $1R2$ and $2R3$. Transitivity requires $1R3$. Therefore add (1, 3).

Question 21

Which relation is an equivalence relation?

- A. Reflexive only
- B. Reflexive and symmetric only
- C. Reflexive, symmetric and transitive
- D. Transitive only

Answer: ✓ C

Explanation:

Equivalence relation requires all three: 1. Reflexive 2. Symmetric 3. Transitive

Question 22

If: $aRb \Leftrightarrow a - b$ is divisible by 3

This relation is:

- A. An equivalence relation
- B. Not reflexive
- C. Not symmetric
- D. Not transitive

Answer: ✓ A

Solution:

Reflexive: $a - a = 0$, which is divisible by 3.

Symmetric: if $a - b$ is divisible by 3, then $b - a$ is also divisible by 3.

Transitive also holds. Therefore it is an equivalence relation.

Question 23

A proof that starts by assuming the opposite of what you want to prove and reaches a contradiction is called:

- A. Direct proof
- B. Proof by contradiction
- C. Mathematical induction
- D. Construction proof

Answer: ✓ B

Question 24

To prove: $n^2 + n$ is even for all integers n , the best first step in induction is:

- A. Assume it is false
- B. Test only $n = 10$
- C. Check the base case
- D. Divide by n

Answer: ✓ C

Explanation:

Mathematical induction begins with: 1. Base case 2. Induction hypothesis 3. Induction step

Question 25

In mathematical induction, assuming the statement is true for $n = k$ is called:

- A. Base case
- B. Inductive hypothesis
- C. Conclusion
- D. Contradiction

Answer: ✓ B

Explanation:

The assumption $P(k)$ is true is the induction hypothesis (inductive hypothesis).

Batch 2: Questions 26-50 (With Answers + Solutions)

Question 26

Let: $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8, 10\}$

Find: $A \cap B$

- A. $\{1, 3, 5\}$
- B. $\{2, 4, 6\}$
- C. $\{8, 10\}$
- D. $\{1, 2, 3, 4, 5, 6, 8, 10\}$

Answer: ✓ B. $\{2, 4, 6\}$

Solution:

Intersection means common elements. Common elements: 2, 4, 6 $\rightarrow A \cap B = \{2, 4, 6\}$

Question 27

If a set has 8 elements, how many elements are in its power set?

- A. 16
- B. 32
- C. 64
- D. 256

Answer: ✓ D. 256

Solution:

Formula: $|P(A)| = 2^n$ where $n = 8$. Therefore $2^8 = 256$

Question 28

If: $|A| = 40$, $|B| = 30$, $|A \cap B| = 10$

Find the number of elements in exactly one of the sets.

- A. 50
- B. 60
- C. 70
- D. 80

Answer: ✓ A. 50

Solution:

Only A: $40 - 10 = 30$. Only B: $30 - 10 = 20$. Exactly one: $30 + 20 = 50$

Question 29

A universal set contains 200 students. 80 offer Java. 90 offer Python. 30 offer both.

How many students offer neither Java nor Python?

- A. 40
- B. 60
- C. 70
- D. 90

Answer: ✓ B. 60

Solution:

Union: $|J \cup P| = 80 + 90 - 30 = 140$. Neither: $200 - 140 = 60$

Question 30

Which statement is TRUE?

- A. Every set is a subset of itself
- B. Empty set contains every set
- C. A subset always has more elements
- D. Two sets cannot have common elements

Answer: ✓ A

Explanation:

Every set satisfies $A \subseteq A$. This is called reflexivity of sets.

Question 31

Simplify: $(A \cup B) \cap A$

- A. A
- B. B
- C. $A \cup B$
- D. $\emptyset$

Answer: ✓ **A**

Solution:

Using absorption law: $A \cap (A \cup B) = A$

Question 32

Simplify: $A \cup (A \cap B)$

- A. B
- B. A
- C. $A \cap B$
- D. $\emptyset$

Answer: ✓ **B**

Explanation:

Absorption law: $A \cup (A \cap B) = A$

Question 33

Which of these is NOT a set operation?

- A. Union
- B. Intersection
- C. Differentiation
- D. Complement

Answer: ✓ **C**

Explanation:

The major set operations include: Union, Intersection, Difference, Complement.

Question 34

If: $A = \{1, 2\}$ and $B = \{a, b, c\}$

How many elements are in: $A \times B$

- A. 5
- B. 6
- C. 8
- D. 9

Answer: ✓ **B**

Solution:

Cartesian product: $|A \times B| = |A| \times |B| = 2 \times 3 = 6$

Question 35

The relation: $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$ on $A = \{1, 2, 3\}$ is:

- A. Reflexive only
- B. Symmetric only
- C. Reflexive and symmetric
- D. Not a relation

Answer: ✓ **C**

Solution:

Reflexive: $(1,1), (2,2), (3,3)$ all exist. Symmetric: $(1,2)$ exists and so does $(2,1)$.

Question 36

A relation is antisymmetric when:

- A. $aRb \Rightarrow bRa$
- B. aRb and $bRa \Rightarrow a = b$
- C. $aRb \Rightarrow aRa$
- D. $aRb \Rightarrow aRc$

Answer: ✓ B

Question 37

Which relation is NOT an equivalence relation?

- A. Equality relation
- B. Congruence modulo n
- C. "Less than or equal to" relation
- D. Same age relation

Answer: ✓ C

Explanation:

$\leq$ is not symmetric. Example: $2 \leq 3$ but $3 \leq 2$ is false.

Question 38

The statement: "If it rains, then the ground is wet" has the form:

- A. $p \vee q$
- B. $p \wedge q$
- C. $p \rightarrow q$
- D. $p \leftrightarrow q$

Answer: ✓ C

Question 39

The negation of: $p \wedge q$ is:

- A. $\neg p \wedge \neg q$
- B. $\neg p \vee \neg q$
- C. $p \vee q$
- D. $p \rightarrow q$

Answer: ✓ B

Solution:

De Morgan's law: $\neg(p \wedge q) = \neg p \vee \neg q$

Question 40

The proposition: $p \vee \neg p$ is:

- A. Contradiction
- B. Tautology
- C. Conditional
- D. False statement

Answer: ✓ B

Explanation:

Either p is true or p is false. The statement is always true (tautology).

Question 41

The proposition: $p \wedge \neg p$ is:

- A. Tautology
- B. Contradiction
- C. Equivalent
- D. Conditional

Answer: ✓ B

Explanation:

A statement cannot be both true and false at the same time (contradiction).

Question 42

How many rows are needed for a truth table containing 4 variables?

- A. 4
- B. 8
- C. 16
- D. 32

Answer: ✓ C

Solution:

Formula: 2^n . $2^4 = 16$

Question 43

The converse of: $p \rightarrow q$ is:

- A. $\neg p \rightarrow \neg q$
- B. $q \rightarrow p$
- C. $\neg q \rightarrow \neg p$
- D. $p \rightarrow \neg q$

Answer: ✓ B

Question 44

The contrapositive of: $p \rightarrow q$ is:

- A. $q \rightarrow p$
- B. $\neg p \rightarrow \neg q$
- C. $\neg q \rightarrow \neg p$
- D. $p \leftrightarrow q$

Answer: ✓ C

Question 45

If: $p = \text{true}$, $q = \text{false}$

Find: $p \wedge q$

- A. True
- B. False
- C. Both
- D. Undefined

Answer: ✓ B

Solution:

AND requires both true. $\text{true} \wedge \text{false} = \text{false}$

Question 46

Which rule of inference states: $p \rightarrow q, p \therefore q$

- A. Modus ponens
- B. Modus tollens
- C. Contradiction
- D. Addition

Answer: ✓ A

Question 47

Which rule states: $p \rightarrow q, \neg q \therefore \neg p$

- A. Modus ponens
- B. Modus tollens
- C. Simplification
- D. Conjunction

Answer: ✓ B

Question 48

A predicate is different from a proposition because:

- A. It has no meaning
- B. It contains variables
- C. It is always false
- D. It cannot be evaluated

Answer: ✓ B

Question 49

The symbol: $\forall$ means:

- A. There exists
- B. For all
- C. Not
- D. Therefore

Answer: ✓ B

Question 50

The symbol: $\exists$ means:

- A. For all
- B. If and only if
- C. There exists
- D. And

Answer: ✓ C

Batch 3: Questions 51-100 (With Answers + Solutions)

Question 51

Construct the truth value of: $p \rightarrow q$ when $p = \text{true}$, $q = \text{false}$

- A. True
- B. False
- C. Both
- D. Cannot determine

Answer: ✓ B. False

Explanation:

An implication $p \rightarrow q$ is false only when p is true and q is false. Therefore $T \rightarrow F = F$

Question 52

The expression: $p \leftrightarrow q$ is true when:

- A. Both have the same truth value
- B. Both are false only
- C. p is true only
- D. q is true only

Answer: ✓ A

Explanation:

Biconditional means “ p if and only if q ”. Both must agree: $T \leftrightarrow T = T$, $F \leftrightarrow F = T$

Question 53

Simplify: $\neg(p \rightarrow q)$

- A. $p \wedge \neg q$
- B. $\neg p \wedge q$
- C. $p \vee q$
- D. $\neg p \vee q$

Answer: ✓ A

Solution:

Recall: $p \rightarrow q \equiv \neg p \vee q$. Negate: $\neg(\neg p \vee q)$. Using De Morgan: $p \wedge \neg q$

Question 54

Which of the following is logically equivalent to: $p \rightarrow q$

- A. $p \wedge q$
- B. $\neg p \vee q$
- C. $p \vee q$
- D. $\neg p \wedge q$

Answer: ✓ B

Question 55

A statement that is true for all possible truth values is called:

- A. Contradiction
- B. Tautology
- C. Predicate
- D. Argument

Answer: ✓ B

Question 56

Find the number of rows required for a truth table with 6 variables.

- A. 6
- B. 12
- C. 32
- D. 64

Answer: ✓ D

Solution:

Formula: 2^n where $n = 6$. Therefore $2^6 = 64$

Question 57

Translate: “Every student passed COS203”

- A. $\exists x (\text{Student}(x) \rightarrow \text{Passed}(x))$
- B. $\forall x (\text{Student}(x) \rightarrow \text{Passed}(x))$
- C. $\forall x (\text{Student}(x) \wedge \text{Passed}(x))$
- D. $\exists x (\text{Passed}(x))$

Answer: ✓ B

Explanation:

“Every” means universal quantifier $\forall$.

Question 58

Translate: “There exists a student who passed COS203”

- A. $\forall x (\text{Passed}(x))$
- B. $\exists x (\text{Passed}(x))$
- C. $\forall x (\text{Student}(x))$
- D. $\text{Student} \rightarrow \text{Passed}$

Answer: ✓ B

Question 59

The negation of: $\forall x P(x)$ is:

- A. $\forall x \neg P(x)$
- B. $\exists x \neg P(x)$
- C. $\exists x P(x)$
- D. $P(x)$

Answer: ✓ B

Explanation:

De Morgan rule for quantifiers: $\neg \forall x P(x) \equiv \exists x \neg P(x)$

Question 60

The negation of: $\exists x P(x)$ is:

- A. $\forall x \neg P(x)$
- B. $\exists x \neg P(x)$
- C. $\forall x P(x)$
- D. $P(x)$

Answer: ✓ A

Question 61

To prove: “the sum of two even numbers is even” the best proof method is:

- A. Direct proof
- B. Contradiction
- C. Induction only
- D. Truth table

Answer: ✓ A

Explanation:

Start by defining $a = 2m$, $b = 2n$. Then $a + b = 2(m + n)$. Therefore even.

Question 62

If: $n = 2k$ then n is:

- A. Odd
- B. Prime
- C. Even
- D. Negative

Answer: ✓ C

Question 63

If: $n = 2k + 1$ then n is:

- A. Even
- B. Odd
- C. Zero
- D. Prime

Answer: ✓ B

Question 64

Prove that the square of an odd number is odd. If $n = 2k + 1$, find n^2 :

- A. $4k^2 + 1$
- B. $4k^2 + 4k + 1$
- C. $2k^2 + 1$
- D. $4k + 1$

Answer: ✓ B

Solution:

$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Therefore it is odd.

Question 65

Mathematical induction is used to prove statements involving:

- A. Only real numbers
- B. Natural numbers
- C. Only matrices
- D. Sets only

Answer: ✓ B

Question 66

The first step in induction is:

- A. Inductive step
- B. Assume false
- C. Base case
- D. Conclusion

Answer: ✓ C

Question 67

In induction, after proving $P(k)$ we prove:

- A. $P(0)$
- B. $P(k + 1)$
- C. $P(k - 1)$
- D. $P(2k)$

Answer: ✓ B

Question 68

For: $1 + 2 + 3 + \dots + n = n(n + 1)/2$ the base case usually checks:

- A. $n = 0$
- B. $n = 1$
- C. $n = 2$
- D. $n = 10$

Answer: ✓ B

Question 69

Proof by contradiction begins by:

- A. Proving the statement directly
- B. Assuming the opposite is true
- C. Using a truth table
- D. Drawing a graph

Answer: ✓ B

Question 70

To prove: $\sqrt{2}$ is irrational, we assume:

- A. $\sqrt{2}$ is rational
- B. $\sqrt{2} = 0$
- C. 2 is rational
- D. No assumption

Answer: ✓ A

Question 71

A proof using: "If Q is false, then P is false" to prove $P \rightarrow Q$ is called:

- A. Direct proof
- B. Contrapositive proof
- C. Induction
- D. Construction

Answer: ✓ B

Question 72

The contrapositive of: $p \rightarrow q$ is:

- A. $q \rightarrow p$
- B. $\neg q \rightarrow \neg p$
- C. $\neg p \rightarrow \neg q$
- D. $p \rightarrow \neg q$

Answer: ✓ B

Question 73

Which statement is always equivalent to the original implication?

- A. Converse
- B. Inverse
- C. Contrapositive
- D. Negation

Answer: ✓ C

Question 74

If a proof reaches: $1 = 0$ the proof has shown:

- A. The statement is true
- B. A contradiction exists
- C. The formula is incomplete
- D. Nothing

Answer: ✓ B

Question 75

Which method is best for proving a formula involving $1 + 2 + 3 + \dots + n$ for all positive integers?

- A. Mathematical induction
- B. Venn diagram
- C. Matrix multiplication
- D. Boolean algebra

Answer: ✓ A

Question 76

Let: $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$

The relationship between A and B is:

- A. $A = B$
- B. $A \subset B$
- C. $B \subset A$
- D. $A \cap B = \emptyset$

Answer: ✓ B. $A \subset B$

Explanation:

Every element of A exists in B, but B contains extra elements. Therefore $A \subset B$

Question 77

If: $A = \{a, b, c\}$ how many proper subsets does A have?

- A. 3
- B. 6
- C. 7
- D. 8

Answer: ✓ C. 7

Solution:

Total subsets: $2^3 = 8$. Proper subsets exclude the set itself: $8 - 1 = 7$

Question 78

A set with exactly one element is called:

- A. Empty set
- B. Singleton set
- C. Universal set
- D. Power set

Answer: ✓ B

Question 79

If: $A = \{1, 2, 3\}$ then $|P(A)|$ is:

- A. 3
- B. 6
- C. 8
- D. 9

Answer: ✓ C

Solution:

$$|P(A)| = 2^n = 2^3 = 8$$

Question 80

Which of the following is TRUE about the empty set?

- A. It has one element
- B. It is a subset of every set
- C. It contains all elements
- D. It cannot be written mathematically

Answer: ✓ B

Explanation:

The empty set satisfies $\emptyset \subseteq A$ for every set A .

Question 81

If: $A \cap B = A$ then:

- A. $A \subseteq B$
- B. $B \subseteq A$
- C. $A = B$
- D. $A \cap B = \emptyset$

Answer: ✓ A

Explanation:

If the intersection gives A , all elements of A must already exist in B .

Question 82

If: $A \cup B = B$ then:

- A. $A \subseteq B$
- B. $B \subseteq A$
- C. A and B are disjoint
- D. A is empty

Answer: ✓ A

Question 83

A relation from A to B contains:

- A. Single numbers
- B. Ordered pairs
- C. Matrices
- D. Boolean values

Answer: ✓ B

Question 84

For a relation to be reflexive on A:

- A. Every element must relate to itself
- B. Every element must relate to another element
- C. No elements should relate
- D. It must be symmetric

Answer: ✓ A

Question 85

A relation that is reflexive, symmetric and transitive is called:

- A. Partial order
- B. Equivalence relation
- C. Function
- D. Cartesian product

Answer: ✓ B

Question 86

The relation “less than or equal to” ($\leq$) is:

- A. Equivalence relation
- B. Partial order
- C. Symmetric relation
- D. Empty relation

Answer: ✓ B

Explanation:

It is reflexive, antisymmetric and transitive. Therefore it is a partial order.

Question 87

Which property is NOT required for a partial order?

- A. Reflexive
- B. Antisymmetric
- C. Transitive
- D. Symmetric

Answer: ✓ D

Question 88

Which of these is a valid direct proof structure?

- A. Assume conclusion is false and stop
- B. Start with assumptions and logically reach conclusion
- C. Use only examples
- D. Draw a graph

Answer: ✓ B

Question 89

Prove: “The sum of two odd numbers is even.” If $a = 2m + 1$ and $b = 2n + 1$, then $a + b =$

- A. $2(m + n) + 1$
- B. $2(m + n + 1)$
- C. $m + n$
- D. $2mn$

Answer: ✓ B

Solution:

$a + b = (2m + 1) + (2n + 1) = 2m + 2n + 2 = 2(m + n + 1)$. Therefore even.

Question 90

The induction step proves:

- A. The first case only
- B. If $P(k)$ is true, then $P(k + 1)$ is true
- C. The statement is false
- D. The variables disappear

Answer: ✓ B

Question 91

In induction, the statement assumed true is called:

- A. Inductive hypothesis
- B. Contradiction
- C. Converse
- D. Negation

Answer: ✓ A

Question 92

Which proof method is commonly used to prove divisibility?

- A. Direct proof
- B. Venn diagram
- C. Matrix method
- D. Truth table

Answer: ✓ A

Question 93

The statement: $(p \rightarrow q) \wedge (q \rightarrow p)$ is equivalent to:

- A. $p \vee q$
- B. $p \leftrightarrow q$
- C. $p \wedge q$
- D. $\neg p$

Answer: ✓ B

Question 94

Find the truth value: $\neg(\text{true} \wedge \text{false})$

- A. True
- B. False
- C. Both
- D. Undefined

Answer: ✓ A

Solution:

First: $\text{true} \wedge \text{false} = \text{false}$. Then: $\neg \text{false} = \text{true}$

Question 95

The expression: $p \vee (p \wedge q)$ simplifies to:

- A. p
- B. q
- C. $p \wedge q$
- D. $p \vee q$

Answer: ✓ A

Explanation:

Absorption law: $p \vee (p \wedge q) = p$

Question 96

A proposition that is always false is called:

- A. Tautology
- B. Contradiction
- C. Equivalence
- D. Predicate

Answer: ✓ B

Question 97

Which symbol represents “if and only if”?

- A. $\rightarrow$
- B. $\wedge$
- C. $\leftrightarrow$
- D. $\neg$

Answer: ✓ C

Question 98

The argument: $p \rightarrow q$, $\neg q$ Therefore: $\neg p$ uses:

- A. Modus ponens
- B. Modus tollens
- C. Addition
- D. Simplification

Answer: ✓ B

Question 99

A statement containing variables whose truth depends on the value of the variable is called:

- A. Proposition
- B. Predicate
- C. Tautology
- D. Contradiction

Answer: ✓ B

Question 100

The main purpose of mathematical induction is to prove:

- A. A statement is true for all natural numbers
- B. A statement is false
- C. A graph exists
- D. A matrix is invertible

Answer: ✓ A

Batch 4: Questions 101-200 — Boolean Algebra, Number Systems, Graphs & Lattices

Question 101

Boolean algebra deals with variables that can have:

- A. Any real value
- B. Only two possible values
- C. Three possible values
- D. Infinite values

Answer: ✓ B

Explanation:

Boolean variables only have 0 (False) and 1 (True).

Question 102

In Boolean algebra: $A + 0 =$

- A. 0
- B. 1
- C. A
- D. A'

Answer: ✓ C

Solution:

Identity law: $A + 0 = A$

Question 103

Simplify: $A \cdot 1$

- A. 0
- B. 1
- C. A
- D. A'

Answer: ✓ C

Explanation:

Identity law: $A \cdot 1 = A$

Question 104

Simplify: $A + A$

- A. A
- B. 0
- C. 1
- D. A'

Answer: ✓ A

Explanation:

Idempotent law: $A + A = A$

Question 105

Simplify: $A \cdot A$

- A. 0
- B. 1
- C. A
- D. A'

Answer: ✓ C

Question 106

Simplify: $A + A'$

- A. A
- B. 0
- C. 1
- D. A'

Answer: ✓ C

Explanation:

Complement law: $A + A' = 1$

Question 107

Simplify: $A \cdot A'$

- A. 1
- B. 0
- C. A
- D. A'

Answer: ✓ B

Explanation:

A variable cannot be both true and false: $A \cdot A' = 0$

Question 108

Which law states: $A + B = B + A$

- A. Associative law
- B. Distributive law
- C. Commutative law
- D. Absorption law

Answer: ✓ C

Question 109

Simplify: $A + AB$

- A. A
- B. B
- C. AB
- D. 0

Answer: ✓ A

Solution:

Absorption law: $A + AB = A$

Question 110

Simplify: $A(A + B)$

- A. B
- B. A
- C. AB
- D. 1

Answer: ✓ B

Solution:

$A(A + B) = AA + AB = A + AB = A$ (absorption)

Question 111

Find the complement of: $(A + B)$

- A. $A' + B'$
- B. $A'B'$
- C. AB
- D. $A + B'$

Answer: ✓ B

Explanation:

De Morgan's law: $(A + B)' = A'B'$

Question 112

Find: $(AB)'$

- A. $A'B'$
- B. $A' + B'$
- C. AB
- D. $A + B$

Answer: ✓ B

Explanation:

De Morgan: $(AB)' = A' + B'$

Question 113

Convert decimal number 10 to binary.

- A. 1010
- B. 1110
- C. 1001
- D. 1100

Answer: ✓ A

Solution:

$10 \div 2 = 5$ r0; $5 \div 2 = 2$ r1; $2 \div 2 = 1$ r0; $1 \div 2 = 0$ r1. Read upward: 1010

Question 114

The binary number: 1111_2 is equal to:

- A. 10
- B. 12
- C. 15
- D. 16

Answer: ✓ C

Solution:

$$1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 8 + 4 + 2 + 1 = 15$$

Question 115

A minterm represents:

- A. A maximum value
- B. A product term containing all variables
- C. A random expression
- D. A graph

Answer: ✓ B

Explanation:

A minterm contains every variable exactly once (e.g. $A'BC$).

Question 116

A Karnaugh map is used for:

- A. Increasing Boolean expression size
- B. Simplifying Boolean expressions
- C. Drawing graphs
- D. Finding determinants

Answer: ✓ B

Question 117

A 3-variable Karnaugh map contains:

- A. 3 cells
- B. 6 cells
- C. 8 cells
- D. 16 cells

Answer: ✓ C

Solution:

Number of cells = $2^n = 2^3 = 8$

Question 118

A 4-variable Karnaugh map contains:

- A. 4 cells
- B. 8 cells
- C. 16 cells
- D. 32 cells

Answer: ✓ C

Question 119

The largest group size allowed in Karnaugh maps is usually:

- A. Any number
- B. Powers of 2
- C. Prime numbers
- D. Odd numbers

Answer: ✓ B

Explanation:

Groups must be powers of 2: 1, 2, 4, 8, 16...

Question 120

In Boolean algebra, multiplication represents:

- A. OR
- B. AND
- C. NOT
- D. XOR

Answer: ✓ B

Question 121

In Boolean algebra, addition represents:

- A. AND
- B. OR
- C. NOT
- D. NAND

Answer: ✓ B

Question 122

The NOT operation is represented by:

- A. +
- B. ·
- C. ´
- D. =

Answer: ✓ C

Explanation:

A´ means NOT A.

Question 123

A lattice is a partially ordered set where:

- A. Every pair has a greatest lower bound and least upper bound
- B. No elements are related
- C. Only one element exists
- D. It has no order

Answer: ✓ A

Question 124

A partially ordered set is also called:

- A. Boolean function
- B. Poset
- C. Matrix
- D. Relation table

Answer: ✓ B

Question 125

A Hasse diagram is used to represent:

- A. Boolean expressions
- B. Partial ordering
- C. Probability
- D. Matrices

Answer: ✓ B

Question 126

Simplify: $A + A'B$

- A. A
- B. $A + B$
- C. AB
- D. B

Answer: ✓ B. $A + B$

Solution:

$$A + A'B = (A + A')(A + B) = 1 \cdot (A + B) = A + B$$

Question 127

Simplify: $AB + A'B$

- A. A
- B. B
- C. AB
- D. 1

Answer: ✓ B

Solution:

Factor B: $B(A + A') = B \cdot 1 = B$

Question 128

Simplify: $(A + B)(A + C)$

- A. $A + BC$
- B. $AB + AC$
- C. A
- D. $B + C$

Answer: ✓ A

Solution:

Expand $\rightarrow A + AC + AB + BC \rightarrow \text{absorb} \rightarrow A + BC$

Question 129

A NAND gate is equivalent to:

- A. AND followed by NOT
- B. OR followed by NOT
- C. NOT followed by AND
- D. XOR gate

Answer: ✓ A

Explanation:

$\text{NAND} = (AB)'$

Question 130

A NOR gate is equivalent to:

- A. AB
- B. $(A + B)'$
- C. $A + B$
- D. A'

Answer: ✓ B

Question 131

The output of an XOR gate is 1 when:

- A. Both inputs are equal
- B. Both inputs are zero
- C. Inputs are different
- D. Both inputs are one

Answer: ✓ C

Question 132

Find: $1011_2 + 0101_2$

- A. 1110
- B. 10000
- C. 1100
- D. 10101

Answer: ✓ B

Solution:

$1011_2 = 11$, $0101_2 = 5$; $11 + 5 = 16 = 10000_2$

Question 133

The duality principle in Boolean algebra means:

- A. Replace + with $\cdot$ and 0 with 1
- B. Remove variables
- C. Add more terms
- D. Convert decimal to binary

Answer: ✓ A

Question 134

A graph consists of:

- A. Vertices and edges
- B. Matrices only
- C. Numbers only
- D. Boolean values

Answer: ✓ A

Question 135

The points in a graph are called:

- A. Edges
- B. Vertices
- C. Paths
- D. Cycles

Answer: ✓ B

Question 136

The lines connecting vertices are called:

- A. Nodes
- B. Edges
- C. Degrees
- D. Trees

Answer: ✓ B

Question 137

The degree of a vertex is:

- A. Number of vertices
- B. Number of edges connected to it
- C. Number of paths
- D. Total graph size

Answer: ✓ B

Question 138

A graph with no loops and no multiple edges is:

- A. Simple graph
- B. Complete graph
- C. Directed graph
- D. Weighted graph

Answer: ✓ A

Question 139

A graph where every pair of vertices is connected is:

- A. Simple graph
- B. Complete graph
- C. Bipartite graph
- D. Tree

Answer: ✓ B

Question 140

A complete graph with n vertices is written as:

- A. K_n
- B. G_n
- C. T_n
- D. P_n

Answer: ✓ A

Question 141

How many edges does K_4 have?

- A. 4
- B. 6
- C. 8
- D. 12

Answer: ✓ B

Solution:

$$E = n(n-1)/2 = 4 \cdot 3/2 = 6$$

Question 142

A graph where edges have direction is:

- A. Simple graph
- B. Directed graph
- C. Complete graph
- D. Tree

Answer: ✓ B

Question 143

A graph where edges have values attached is:

- A. Weighted graph
- B. Null graph
- C. Complete graph
- D. Empty graph

Answer: ✓ A

Question 144

A path that starts and ends at the same vertex is:

- A. Tree
- B. Cycle
- C. Edge
- D. Degree

Answer: ✓ B

Question 145

A graph with n vertices and no cycles but connected is:

- A. Tree
- B. Complete graph
- C. Bipartite graph
- D. Directed graph

Answer: ✓ A

Question 146

A tree with n vertices contains:

- A. n edges
- B. $n+1$ edges
- C. $n-1$ edges
- D. $2n$ edges

Answer: ✓ C

Question 147

A connected graph has an Euler circuit if:

- A. All vertices have even degree
- B. All vertices have odd degree
- C. It has no edges
- D. It has one vertex

Answer: ✓ A

Question 148

An Euler path uses:

- A. Every vertex once
- B. Every edge once
- C. Every cycle twice
- D. No edges

Answer: ✓ B

Question 149

A Hamiltonian path visits:

- A. Every edge once
- B. Every vertex once
- C. Only one vertex
- D. No vertices

Answer: ✓ B

Question 150

The adjacency matrix of a graph represents:

- A. Relationship between vertices
- B. Boolean laws
- C. Probability values only
- D. Set operations

Answer: ✓ A

Question 151

A graph with all vertices having even degree has:

- A. Euler circuit
- B. Hamiltonian circuit only
- C. No path
- D. No edges

Answer: ✓ A

Explanation:

A connected graph has an Euler circuit when every vertex has even degree.

Question 152

A graph with exactly two odd-degree vertices has:

- A. Euler circuit
- B. Euler path
- C. Hamiltonian cycle
- D. Complete graph

Answer: ✓ B

Explanation:

0 odd vertices → Euler circuit; exactly 2 odd vertices → Euler path.

Question 153

Which of the following is NOT a requirement for an Euler circuit?

- A. Connected graph
- B. All vertices have even degree
- C. Every edge used once
- D. Every vertex has degree 2

Answer: ✓ D

Explanation:

Vertices only need even degree, not specifically degree 2.

Question 154

A graph is called bipartite if:

- A. It has no vertices
- B. Vertices can be divided into two sets with edges only between sets
- C. Every vertex connects to every other vertex
- D. It contains a cycle

Answer: ✓ B

Question 155

A complete bipartite graph is represented as:

- A. K_n
- B. $K_{m,n}$
- C. G_n
- D. T_n

Answer: ✓ B

Question 156

How many edges are in $K_{3,4}$?

- A. 7
- B. 12
- C. 16
- D. 24

Answer: ✓ B

Solution:

$K_{m,n}$ has $m \cdot n$ edges: $3 \times 4 = 12$

Question 157

A graph with no cycles and connected is called:

- A. Tree
- B. Circuit
- C. Complete graph
- D. Multigraph

Answer: ✓ A

Question 158

A spanning tree of a graph contains:

- A. All vertices and minimum edges needed to connect them
- B. All possible edges
- C. Only one vertex
- D. No edges

Answer: ✓ A

Question 159

A tree with 10 vertices has how many edges?

- A. 8
- B. 9
- C. 10
- D. 11

Answer: ✓ B

Solution:

$$E = V - 1 = 10 - 1 = 9$$

Question 160

A graph represented using rows and columns showing connections is:

- A. Truth table
- B. Adjacency matrix
- C. Karnaugh map
- D. Hasse diagram

Answer: ✓ B

Question 161

For an undirected graph, the adjacency matrix is usually:

- A. Symmetric
- B. Always zero
- C. Negative
- D. Diagonal only

Answer: ✓ A

Explanation:

If A connects to B, then B connects to A.

Question 162

The sum of degrees of all vertices in a graph equals:

- A. Number of vertices
- B. Twice the number of edges
- C. Number of edges
- D. Zero

Answer: ✓ B

Solution:

Handshaking theorem: $\sum \text{degree} = 2E$

Question 163

A graph has 5 vertices with degrees 2, 3, 4, 1, 2. How many edges?

- A. 5
- B. 6
- C. 12
- D. 24

Answer: ✓ B

Solution:

Sum degrees = 12 $\rightarrow 2E = 12 \rightarrow E = 6$

Question 164

A Hamiltonian circuit:

- A. Uses every edge once
- B. Visits every vertex once and returns to start
- C. Has no vertices
- D. Must have all even degrees

Answer: ✓ B

Question 165

Which statement is TRUE?

- A. Every Euler graph is Hamiltonian
- B. Every Hamiltonian graph is Eulerian
- C. They are different concepts
- D. They are identical

Answer: ✓ C

Explanation:

Euler focuses on edges; Hamiltonian focuses on vertices.

Question 166

A graph with one vertex and no edges is called:

- A. Complete graph
- B. Null graph
- C. Tree only
- D. Bipartite graph

Answer: ✓ B

Question 167

The degree of a vertex with a loop is counted as:

- A. 0
- B. 1
- C. 2
- D. 3

Answer: ✓ C

Explanation:

A loop contributes two to the degree.

Question 168

The maximum number of edges in a simple graph with n vertices is:

- A. n^2
- B. $n(n-1)/2$
- C. $2n$
- D. $n-1$

Answer: ✓ B

Question 169

Find maximum edges in a simple graph with 6 vertices.

- A. 10
- B. 12
- C. 15
- D. 20

Answer: ✓ C

Solution:

$$n(n-1)/2 = 6 \cdot 5/2 = 15$$

Question 170

A graph with every vertex connected to every other vertex is:

- A. Complete graph
- B. Tree
- C. Bipartite graph
- D. Path graph

Answer: ✓ A

Question 171

A Hasse diagram removes:

- A. Vertices
- B. Loops and unnecessary edges
- C. All relationships
- D. Labels

Answer: ✓ B

Question 172

In a partially ordered set, if $a \leq b$ and $b \leq a$ then:

- A. $a = b$
- B. $a > b$
- C. They are unrelated
- D. Impossible

Answer: ✓ A

Explanation:

This is antisymmetry.

Question 173

The least upper bound is also called:

- A. Meet
- B. Join
- C. Edge
- D. Degree

Answer: ✓ B

Question 174

The greatest lower bound is also called:

- A. Join
- B. Meet
- C. Path
- D. Circuit

Answer: ✓ B

Question 175

A Boolean algebra is a special type of:

- A. Graph
- B. Lattice
- C. Matrix
- D. Tree

Answer: ✓ B

Question 176

A graph with no cycles and containing all vertices connected is:

- A. Complete graph
- B. Tree
- C. Multigraph
- D. Directed graph

Answer: ✓ B. Tree

Explanation:

A tree must be connected and have no cycles.

Question 177

A tree with 15 vertices contains:

- A. 14 edges
- B. 15 edges
- C. 16 edges
- D. 30 edges

Answer: ✓ A. 14 edges

Solution:

$$E = V - 1 = 15 - 1 = 14$$

Question 178

A connected graph has an Euler circuit when:

- A. Exactly two vertices have odd degree
- B. All vertices have even degree
- C. All vertices have odd degree
- D. It has no edges

Answer: ✓ B

Question 179

The degree sequence 3, 3, 2, 2 has total degree:

- A. 5
- B. 8
- C. 10
- D. 12

Answer: ✓ C. 10

Solution:

$$3+3+2+2 = 10. \text{ Then } 2E = 10 \rightarrow E = 5.$$

Question 180

If the sum of degrees in a graph is 20, the number of edges is:

- A. 5
- B. 10
- C. 20
- D. 40

Answer: ✓ B

Solution:

$$2E = 20 \rightarrow E = 10$$

Question 181

Which graph has every vertex connected to every other vertex?

- A. Tree
- B. Complete graph
- C. Path graph
- D. Null graph

Answer: ✓ B

Question 182

The number of edges in K_5 is:

- A. 5
- B. 10
- C. 15
- D. 20

Answer: ✓ B

Solution:

$$E = n(n-1)/2 = 5 \cdot 4/2 = 10$$

Question 183

Which graph cannot contain an Euler circuit?

- A. Connected graph with all even degrees
- B. Graph with odd degree vertices
- C. Complete graph with even degrees
- D. Cycle graph

Answer: ✓ B

Explanation:

Odd degree vertices prevent Euler circuits.

Question 184

A path that visits every vertex exactly once is:

- A. Euler path
- B. Hamiltonian path
- C. Circuit
- D. Tree

Answer: ✓ B

Question 185

The adjacency matrix of a graph with 5 vertices has size:

- A. 5×5
- B. 5×1
- C. 10×10
- D. 2×5

Answer: ✓ A

Explanation:

For n vertices: $n \times n$ matrix.

Question 186

In an adjacency matrix, a 1 means:

- A. No connection
- B. A connection exists
- C. Vertex removed
- D. Graph is complete

Answer: ✓ B

Question 187

A graph is planar if:

- A. It has no vertices
- B. It can be drawn without edge crossings
- C. It has only one edge
- D. It is complete

Answer: ✓ B

Question 188

Euler's formula for connected planar graphs is:

- A. $V + E + F = 2$
- B. $V - E + F = 2$
- C. $V - E - F = 2$
- D. $V + E - F = 0$

Answer: ✓ B

Question 189

In Euler's planar formula, F represents:

- A. Faces
- B. Functions
- C. Fields
- D. Factors

Answer: ✓ A

Question 190

A lattice requires every pair of elements to have:

- A. Only one relation
- B. Greatest lower bound and least upper bound
- C. No ordering
- D. Equal values

Answer: ✓ B

Question 191

In lattice theory, the symbol $\wedge$ represents:

- A. Join
- B. Meet
- C. Addition
- D. Multiplication

Answer: ✓ B

Question 192

In lattice theory, the symbol $\vee$ represents:

- A. Join
- B. Meet
- C. Complement
- D. Negation

Answer: ✓ A

Question 193

A Boolean lattice contains:

- A. Complement for every element
- B. Only one element
- C. No order
- D. No operations

Answer: ✓ A

Question 194

Simplify: $A + AB + A'C$

- A. $A + C$
- B. AB
- C. $B + C$
- D. $A + B$

Answer: ✓ A

Solution:

$$A + AB = A \rightarrow A + A'C = A + C$$

Question 195

Simplify: $AB + A'B + AB'$

- A. $A + B$
- B. AB
- C. B
- D. 0

Answer: ✓ A

Solution:

$$AB + A'B = B \rightarrow B + AB' = A + B$$

Question 196

A Boolean expression in which every term is ANDed is called:

- A. Sum of products
- B. Product of sums
- C. Minterm only
- D. Maximum term

Answer: ✓ A

Question 197

A Boolean expression in which terms are ORed together is:

- A. Product of sums
- B. Sum of products
- C. Matrix form
- D. Graph form

Answer: ✓ B

Question 198

A maxterm is:

- A. Product term containing all variables
- B. Sum term containing all variables
- C. A graph edge
- D. A matrix element

Answer: ✓ B

Question 199

The purpose of Boolean simplification is to:

- A. Increase circuit complexity
- B. Reduce hardware requirements
- C. Remove all variables
- D. Create graphs

Answer: ✓ B

Question 200

The main purpose of a Karnaugh map is:

- A. Solve differential equations
- B. Minimize Boolean expressions
- C. Find graph paths
- D. Calculate probability

Answer: ✓ B

Batch 5: Questions 201-300 — Matrices, Counting, Probability & Recurrence

Question 201

A matrix with m rows and n columns has order:

- A. $m + n$
- B. $m \times n$
- C. $n - m$
- D. m^n

Answer: ✓ B. $m \times n$

Explanation:

Matrix order is always Rows $\times$ Columns. Example: a 2×3 matrix has 2 rows and 3 columns.

Question 202

The matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is a:

- A. Row matrix
- B. Column matrix
- C. Square matrix
- D. Null matrix

Answer: ✓ C

Explanation:

Number of rows = number of columns (2×2), therefore it is square.

Question 203

A matrix with all entries equal to zero is called:

- A. Identity matrix
- B. Zero matrix
- C. Diagonal matrix
- D. Scalar matrix

Answer: ✓ B

Question 204

The identity matrix of order 3 is:

- A. All entries 1
- B. 1s on main diagonal, 0s elsewhere
- C. All entries 0
- D. 3s on main diagonal

Answer: ✓ B

Explanation:

Identity matrix has 1 on the main diagonal and 0 elsewhere.

Question 205

Two matrices can be added only when they have:

- A. Same determinant
- B. Same order
- C. Same diagonal
- D. Same values

Answer: ✓ B

Question 206

Find: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

- A. $\begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$
- B. $\begin{bmatrix} 5 & 12 \\ 21 & 32 \end{bmatrix}$
- C. $\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$
- D. None

Answer: ✓ A

Solution:

Add corresponding elements: $1+5=6$, $2+6=8$, $3+7=10$, $4+8=12$.

Question 207

Matrix multiplication is possible when:

- A. Rows are equal
- B. Columns of first matrix equal rows of second
- C. Both matrices are square
- D. Both matrices are identical

Answer: ✓ B

Question 208

Find the determinant of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

- A. $ab - cd$
- B. $ad - bc$
- C. $ac - bd$
- D. $a + b + c + d$

Answer: ✓ B

Formula:

$$|A| = ad - bc$$

Question 209

Find determinant: $|[[2, 3], [4, 5]]|$

- A. -2
- B. 2
- C. 10
- D. 23

Answer: ✓ A

Solution:

$$(2)(5) - (3)(4) = 10 - 12 = -2$$

Question 210

A matrix is invertible if:

- A. Determinant is zero
- B. Determinant is not zero
- C. All elements are zero
- D. It has one row

Answer: ✓ B

Question 211

The inverse of a matrix exists only for:

- A. Singular matrices
- B. Square matrices with non-zero determinant
- C. Zero matrices
- D. Row matrices

Answer: ✓ B

Question 212

The transpose of $[[1, 2], [3, 4]]$ is:

- A. $[[1, 3], [2, 4]]$
- B. $[[4, 3], [2, 1]]$
- C. $[[1, 2], [3, 4]]$
- D. None

Answer: ✓ A

Explanation:

Rows become columns.

Question 213

A determinant of zero means the matrix is:

- A. Identity
- B. Singular
- C. Invertible
- D. Diagonal

Answer: ✓ B

Question 214

The number of ways of arranging n objects is:

- A. n^2
- B. $n!$
- C. 2^n
- D. $n + 1$

Answer: ✓ B

Question 215

Calculate: $5!$

- A. 20
- B. 60
- C. 120
- D. 240

Answer: ✓ C

Solution:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Question 216

The number of ways to arrange 3 different books is:

- A. 3
- B. 6
- C. 9
- D. 12

Answer: ✓ B

Solution:

$$3! = 3 \times 2 \times 1 = 6$$

Question 217

Permutation is concerned mainly with:

- A. Selection only
- B. Arrangement
- C. Probability only
- D. Addition

Answer: ✓ B

Question 218

Combination is concerned mainly with:

- A. Arrangement
- B. Selection without order
- C. Multiplication
- D. Graphs

Answer: ✓ B

Question 219

The formula for combinations is:

- A. $n! / r!$
- B. $n! / (r!(n - r)!)$
- C. n^r
- D. r^n

Answer: ✓ B

Question 220

Find: 5C_2

- A. 5
- B. 10
- C. 20
- D. 25

Answer: ✓ B

Solution:

$${}^5C_2 = 5! / (2! 3!) = (5 \times 4) / 2 = 10$$

Question 221

Find: 6P_2

- A. 12
- B. 20
- C. 30
- D. 36

Answer: ✓ C

Solution:

$${}^nP_r = n! / (n-r)! \rightarrow 6 \times 5 = 30$$

Question 222

The multiplication principle states:

- A. Add possibilities always
- B. Multiply independent choices
- C. Divide all choices
- D. Ignore choices

Answer: ✓ B

Question 223

If a task can be done in 3 ways and another in 4 ways, total ways:

- A. 7
- B. 12
- C. 16
- D. 1

Answer: ✓ B

Solution:

$$3 \times 4 = 12$$

Question 224

The pigeonhole principle states that:

- A. Objects cannot repeat
- B. More objects than boxes means at least one box has multiple objects
- C. All boxes must be equal
- D. There are no limits

Answer: ✓ B

Question 225

If 10 students are placed into 9 rooms, at least one room contains:

- A. No student
- B. Exactly one student
- C. More than one student
- D. Ten students

Answer: ✓ C

Explanation:

More objects than containers means repetition must occur.

Question 226

The inclusion-exclusion principle is used to find:

- A. Matrix inverse
- B. Number of elements in overlapping sets
- C. Graph degree
- D. Boolean complement

Answer: ✓ B

Explanation:

It corrects double counting when sets overlap.

Question 227

For two sets A and B: $|A \cup B| =$

- A. $|A| + |B|$
- B. $|A| + |B| - |A \cap B|$
- C. $|A| - |B|$
- D. $|A \cap B|$

Answer: ✓ B

Question 228

If $|A| = 20$, $|B| = 15$, $|A \cap B| = 5$, find $|A \cup B|$

- A. 25
- B. 30
- C. 40
- D. 10

Answer: ✓ B. 30

Solution:

$$|A \cup B| = 20 + 15 - 5 = 30$$

Question 229

The probability of an impossible event is:

- A. 0
- B. 1
- C. 0.5
- D. -1

Answer: ✓ A

Question 230

The probability of a certain event is:

- A. 0
- B. 0.25
- C. 1
- D. 2

Answer: ✓ C

Question 231

Probability is calculated as:

- A. Total / Favourable
- B. Favourable / Total
- C. Favourable + Total
- D. Favourable – Total

Answer: ✓ B

Question 232

A fair coin is tossed once. Probability of getting Head:

- A. 1
- B. $1/2$
- C. $1/3$
- D. 0

Answer: ✓ B

Solution:

Possible outcomes: H, T. Favourable: 1 → $1/2$

Question 233

A die is rolled once. Probability of getting 6:

- A. $1/2$
- B. $1/3$
- C. $1/6$
- D. 6

Answer: ✓ C

Solution:

One favourable outcome out of 6 → $1/6$

Question 234

Two dice are rolled. Total possible outcomes are:

- A. 6
- B. 12
- C. 36
- D. 18

Answer: ✓ C

Solution:

$$6 \times 6 = 36$$

Question 235

Conditional probability is written as:

- A. $P(A + B)$
- B. $P(A | B)$
- C. $P(A - B)$
- D. $P(A \times B)$

Answer: ✓ B

Question 236

The formula for conditional probability is:

- A. $P(A|B) = P(A \cap B) / P(B)$
- B. $P(A) + P(B)$
- C. $P(A) - P(B)$
- D. $P(A)P(B)$

Answer: ✓ A

Question 237

If two events cannot happen together, they are:

- A. Independent
- B. Mutually exclusive
- C. Equal
- D. Certain

Answer: ✓ B

Question 238

For mutually exclusive events: $P(A \cap B) =$

- A. 1
- B. 0
- C. 2
- D. $P(A)$

Answer: ✓ B

Question 239

Two events are independent if:

- A. $P(A \cap B) = P(A)P(B)$
- B. $P(A) = P(B)$
- C. $P(A) = 0$
- D. $P(A \cap B) = 1$

Answer: ✓ A

Question 240

Bayes theorem is used to calculate:

- A. Matrix multiplication
- B. Conditional probability
- C. Graph paths
- D. Boolean expressions

Answer: ✓ B

Question 241

The complement rule states:

- A. $P(A') = 1 - P(A)$
- B. $P(A') = P(A)$
- C. $P(A') = 0$
- D. $P(A') = 2P(A)$

Answer: ✓ A

Question 242

If $P(A) = 0.7$ then $P(A')$ is:

- A. 0.7
- B. 1.7
- C. 0.3
- D. 0

Answer: ✓ C

Solution:

$$P(A') = 1 - 0.7 = 0.3$$

Question 243

A random variable assigns:

- A. Numbers to outcomes
- B. Graphs to sets
- C. Matrices to equations
- D. Variables to matrices

Answer: ✓ A

Question 244

A probability distribution must have total probability:

- A. 0
- B. 1
- C. 2
- D. 100

Answer: ✓ B

Question 245

A binomial experiment has:

- A. Unlimited outcomes
- B. Two possible outcomes
- C. No probability
- D. Only one trial

Answer: ✓ B

Question 246

The binomial formula is:

- A. $P(X=x) = {}^nC_x p^x(1-p)^{n-x}$
- B. $P = x + n$
- C. $P = p + n$
- D. $P = n!$

Answer: ✓ A

Question 247

In binomial distribution, n represents:

- A. Probability
- B. Number of trials
- C. Success rate
- D. Failure rate

Answer: ✓ B

Question 248

In binomial distribution, p represents:

- A. Probability of success
- B. Number of trials
- C. Total outcomes
- D. Failure only

Answer: ✓ A

Question 249

The expected value of a random variable is also called:

- A. Mean
- B. Median
- C. Mode
- D. Range

Answer: ✓ A

Question 250

A generating function is mainly used to:

- A. Represent sequences algebraically
- B. Draw graphs
- C. Find determinants
- D. Simplify Boolean expressions

Answer: ✓ A

Question 251

If two events A and B are independent, then:

- A. $P(A \cap B) = P(A) + P(B)$
- B. $P(A \cap B) = P(A)P(B)$
- C. $P(A \cap B) = 0$
- D. $P(A \cap B) = 1$

Answer: ✓ B

Explanation:

Independent events: $P(A \cap B) = P(A)P(B)$

Question 252

A bag contains 5 red and 3 blue balls. Probability of selecting a red ball:

- A. $3/8$
- B. $5/8$
- C. $1/5$
- D. $8/5$

Answer: ✓ B

Solution:

Total = 8, Red = 5 $\rightarrow 5/8$

Question 253

A card is selected from a standard deck of 52 cards. Probability of getting an Ace:

- A. $1/52$
- B. $4/52$
- C. $13/52$
- D. $26/52$

Answer: ✓ B

Solution:

4 Aces out of 52 $\rightarrow 4/52$

Question 254

If $P(A)=0.4$, $P(B)=0.5$ and independent, find $P(A \cap B)$

- A. 0.9
- B. 0.2
- C. 0.1
- D. 0.5

Answer: ✓ B

Solution:

$0.4 \times 0.5 = 0.2$

Question 255

A probability value must always lie between:

- A. -1 and 1
- B. 0 and 1
- C. 1 and 10
- D. -10 and 10

Answer: ✓ B

Question 256

The probability of the complement of an event is:

- A. $1 - P(A)$
- B. $P(A) + 1$
- C. $P(A)^2$
- D. $2P(A)$

Answer: ✓ A

Question 257

Bayes theorem is expressed as:

- A. $P(A|B) = [P(B|A)P(A)] / P(B)$
- B. $P(A+B) = P(A)P(B)$
- C. $P(A) = P(B)$
- D. $P(A) = 1 - P(B)$

Answer: ✓ A

Question 258

A box contains 4 white and 6 black balls. Two balls selected without replacement. Probability changes because:

- A. Events are independent
- B. Events are dependent
- C. There is no probability
- D. Both are impossible

Answer: ✓ B

Explanation:

Removing the first ball changes the second selection.

Question 259

The number of ways to choose 2 students from 5 students is:

- A. 5
- B. 10
- C. 20
- D. 25

Answer: ✓ B

Solution:

$${}^5C_2 = 10$$

Question 260

The number of ways to arrange 4 different objects is:

- A. 4
- B. 8
- C. 24
- D. 16

Answer: ✓ C

Solution:

$$4! = 24$$

Question 261

A recurrence relation defines:

- A. A term using previous terms
- B. A graph edge
- C. A matrix inverse
- D. A Boolean law

Answer: ✓ A

Question 262

The Fibonacci sequence follows:

- A. $F_n = F_{n-1} + F_{n-2}$
- B. $F_n = n^{n-1} + F_{n-2}$
- C. $F_n = 2n$
- D. $F_n = n!$

Answer: ✓ A

Question 263

The first two Fibonacci numbers are usually:

- A. 1, 1
- B. 0, 1
- C. 2, 3
- D. 5, 8

Answer: ✓ B

Question 264

A recurrence relation requires:

- A. Initial conditions
- B. No starting value
- C. Only graphs
- D. Matrices only

Answer: ✓ A

Question 265

Solve: $a_n = 2a_{n-1}$, $a_0 = 1$. Find a_3 .

- A. 2
- B. 4
- C. 8
- D. 16

Answer: ✓ C

Solution:

$$a_1=2, a_2=4, a_3=8$$

Question 266

A generating function converts a sequence into:

- A. A polynomial-like expression
- B. A graph
- C. A matrix
- D. A set

Answer: ✓ A

Question 267

The generating function for 1,1,1,1,... is:

- A. $1/(1 - x)$
- B. $1 + x$
- C. x^2
- D. $1 - x$

Answer: ✓ A

Question 268

The coefficient of x^n in a generating function represents:

- A. A sequence term
- B. A matrix element
- C. A graph vertex
- D. A probability only

Answer: ✓ A

Question 269

The purpose of recurrence relations is to:

- A. Describe sequences step-by-step
- B. Draw circles
- C. Find determinants
- D. Simplify logic gates

Answer: ✓ A

Question 270

A discrete probability distribution deals with:

- A. Countable outcomes
- B. Continuous values only
- C. Graphs only
- D. Matrices only

Answer: ✓ A

Question 271

The expected value formula is:

- A. $E(X) = \sum x P(x)$
- B. $E(X) = x + P(x)$
- C. $E(X) = P(x)/x$
- D. $E(X) = x^2$

Answer: ✓ A

Question 272

Variance measures:

- A. Average value only
- B. Spread of data
- C. Number of outcomes
- D. Probability of success only

Answer: ✓ B

Question 273

Standard deviation is:

- A. Square root of variance
- B. Twice variance
- C. Variance squared
- D. Probability

Answer: ✓ A

Question 274

A random experiment is one where:

- A. Outcome is known before performing
- B. Outcome cannot be predicted exactly
- C. There is no outcome
- D. Only one result exists

Answer: ✓ B

Question 275

The sample space contains:

- A. All possible outcomes
- B. Only successful outcomes
- C. No outcomes
- D. Probability values

Answer: ✓ A

Question 276

A sample space with 8 equally likely outcomes has probability of one outcome as:

- A. $1/8$
- B. 8
- C. $1/4$
- D. 0

Answer: ✓ A

Solution:

$$P = 1/8$$

Question 277

If $P(A) = 0.6$ then $P(A')$ is:

- A. 0.6
- B. 0.4
- C. 1.6
- D. 0

Answer: ✓ B

Solution:

$$1 - 0.6 = 0.4$$

Question 278

The probability of getting an even number on a fair die is:

- A. $1/6$
- B. $1/2$
- C. $2/3$
- D. 1

Answer: ✓ B

Solution:

$$\text{Even: } 2, 4, 6 \rightarrow 3/6 = 1/2$$

Question 279

If events A and B are mutually exclusive: $P(A \cup B) =$

- A. $P(A) + P(B)$
- B. $P(A)P(B)$
- C. $P(A) - P(B)$
- D. 0

Answer: ✓ A

Question 280

A probability distribution must satisfy:

- A. $\sum P(x) = 1$
- B. $\sum P(x) = 0$
- C. $\sum P(x) = 2$
- D. $P(x) < 0$

Answer: ✓ A

Question 281

Find: 7C_3

- A. 21
- B. 35
- C. 42
- D. 56

Answer: ✓ B

Solution:

$$(7 \times 6 \times 5) / (3 \times 2 \times 1) = 35$$

Question 282

Find: 7P_3

- A. 35
- B. 210
- C. 343
- D. 21

Answer: ✓ B

Solution:

$$7 \times 6 \times 5 = 210$$

Question 283

The difference between permutation and combination is:

- A. Combination considers order
- B. Permutation considers order
- C. Both are identical
- D. Neither uses factorials

Answer: ✓ B

Question 284

How many binary strings of length 4 exist?

- A. 4
- B. 8
- C. 16
- D. 32

Answer: ✓ C

Solution:

$$2^4 = 16$$

Question 285

The pigeonhole principle is useful when:

- A. Proving repetition must occur
- B. Solving matrices
- C. Finding determinants
- D. Simplifying Boolean equations

Answer: ✓ A

Question 286

If 13 people are in a room, at least two must have the same birth month because:

- A. There are 12 months
- B. There are 13 months
- C. Everyone has different months
- D. Probability is zero

Answer: ✓ A

Explanation:

13 people > 12 months → repetition must happen.

Question 287

Recurrence: $a_n = a_{n-1} + 2$, $a_1 = 3$. Find a_4 .

- A. 5
- B. 7
- C. 9
- D. 11

Answer: ✓ C

Solution:

$a_2=5$, $a_3=7$, $a_4=9$

Question 288

The recurrence $a_n = 3a_{n-1}$ represents:

- A. Arithmetic sequence
- B. Geometric sequence
- C. Random sequence
- D. Boolean sequence

Answer: ✓ B

Question 289

The common ratio of 2, 6, 18, 54 is:

- A. 2
- B. 3
- C. 6
- D. 18

Answer: ✓ B

Question 290

A sequence 1, 3, 5, 7, ... is:

- A. Geometric
- B. Arithmetic
- C. Random
- D. Constant

Answer: ✓ B

Explanation:

Common difference = +2

Question 291

In $G(x) = a_0 + a_1x + a_2x^2 + \dots$, the coefficient a_2 represents:

- A. Constant term
- B. Coefficient of x^2
- C. Power of x only
- D. The variable

Answer: ✓ B

Question 292

The generating function $1 + x + x^2 + x^3 + \dots$ equals:

- A. $1/(1 - x)$
- B. $1 - x$
- C. x^2
- D. $1/x$

Answer: ✓ A

Question 293

The determinant of an identity matrix is:

- A. 0
- B. 1
- C. -1
- D. Depends on size

Answer: ✓ B

Question 294

If two rows of a determinant are identical, the determinant is:

- A. 1
- B. 0
- C. 2
- D. Undefined

Answer: ✓ B

Question 295

A singular matrix has determinant:

- A. 1
- B. 0
- C. -1
- D. Greater than 1

Answer: ✓ B

Question 296

The inverse of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is:

- A. $(1/(ad-bc)) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
- B. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
- C. Zero matrix
- D. Identity matrix

Answer: ✓ A

Question 297

Matrix multiplication is generally:

- A. Commutative
- B. Not always commutative
- C. Always zero
- D. Impossible

Answer: ✓ B

Explanation:

Usually $AB \neq BA$

Question 298

A diagonal matrix contains non-zero values:

- A. Anywhere
- B. Only on the main diagonal
- C. Only outside diagonal
- D. Nowhere

Answer: ✓ B

Question 299

The main diagonal of a matrix contains elements where:

- A. Row number equals column number
- B. Row is zero
- C. Column is zero
- D. Values are negative

Answer: ✓ A

Question 300

The major goal of discrete mathematics in computer science is:

- A. Understanding structures used in computing
- B. Learning only arithmetic
- C. Avoiding logic
- D. Removing algorithms

Answer: ✓ A

End of Questions 1–300

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